

ALEXANDRIA, Romania

WMO: 154890

Lat: 43.9781N

Lon: 25.3528E

Elev: 252

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	15.1	7.2	8.0	11.3	12.2	10.3	15.3	32.1	34.2	26.3	31.4	3.5	270	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.9	94.4	70.9	91.8	69.6	89.2	68.9	73.6	88.7	72.3	86.4	71.0	84.5	6.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.2	108.4	78.1	67.9	103.5	77.3	66.6	99.0	76.0	37.3	88.7	36.1	86.9	35.0	84.6	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	18.8	16.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	28.9	37.6	45.8	53.1	63.6	72.1	75.0	75.5	66.6	54.0	42.4	34.4		
	DBStd	17.27	8.83	8.61	7.94	7.50	5.66	5.40	4.56	5.03	6.68	7.92	8.15	7.44		
	HDD50	2023	653	351	176	54	1	0	0	0	0	51	254	483		
	HDD65	4979	1118	766	597	361	94	8	0	3	62	344	678	948		
	CDD50	3543	0	5	44	146	421	662	775	791	497	176	26	0		
	CDD65	1025	0	0	0	3	49	219	311	329	109	5	0	0		
	CDH74	10969	0	0	9	99	555	2151	3261	3650	1154	89	1	0		
	CDH80	4636	0	0	0	14	117	874	1452	1761	406	12	0	0		
	WSAvg	6.1	6.0	6.7	6.9	7.3	6.3	5.4	5.3	5.8	6.0	5.8	5.7	6.1		
	PrecAvg	21.4	1.5	1.1	1.3	1.9	2.4	2.6	2.2	1.4	2.0	1.9	1.3	1.7		
(15) Precipitation	PrecMax	34.1	4.1	3.1	3.1	3.7	5.4	5.8	7.0	5.2	5.8	4.3	3.3	6.0		
	PrecMin	12.3	0.1	0.1	0.0	0.6	0.6	0.4	0.4	0.1	0.2	0.0	0.0	0.1		
	PrecStd	4.9	0.9	0.7	0.8	0.8	1.1	1.3	1.5	1.2	1.4	1.3	0.9	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	65.3	75.1	82.0	87.7	96.7	98.6	98.1	92.9	82.6	70.2	56.0		
		MCWB	46.1	55.5	57.6	63.6	69.3	71.5	74.3	71.1	68.2	63.6	58.3	49.3		
	2%	DB	46.5	58.6	68.8	76.6	83.5	91.7	93.8	94.5	88.4	76.8	62.9	51.5		
		MCWB	42.4	51.2	54.2	61.5	65.9	70.7	71.3	70.7	65.9	60.8	55.5	46.8		
	5%	DB	42.8	53.8	64.3	71.3	80.4	88.1	91.1	91.8	84.3	72.3	57.8	47.2		
		MCWB	40.0	48.2	52.0	57.9	64.6	70.3	69.9	69.2	64.2	59.0	52.6	43.8		
	10%	DB	39.7	49.5	59.6	67.2	76.8	85.2	88.0	89.5	80.5	67.7	53.5	43.5		
		MCWB	37.9	45.3	49.4	55.7	62.9	68.8	69.1	68.6	63.8	56.9	50.4	41.3		
	0.4%	WB	46.6	56.3	58.3	64.9	70.5	75.5	74.7	75.5	71.2	64.9	59.3	50.1		
		MCDB	52.1	64.5	72.8	79.3	84.3	90.0	93.2	93.5	87.3	76.8	68.3	55.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.8	51.8	55.2	62.4	67.9	73.4	73.3	73.4	68.1	62.3	56.4	47.3		
		MCDB	45.9	57.6	66.7	76.0	79.7	86.4	88.4	89.2	81.3	72.3	63.0	51.7		
	5%	WB	40.4	48.8	52.5	59.8	65.8	71.9	72.0	71.6	66.3	60.3	54.0	44.5		
		MCDB	42.2	53.6	63.1	70.2	77.3	85.0	85.9	86.1	79.8	70.5	58.0	47.1		
	10%	WB	37.9	45.9	50.1	57.1	64.0	70.3	70.9	70.2	64.9	58.2	51.4	41.7		
		MCDB	39.4	49.5	58.6	65.3	74.7	82.4	83.6	84.7	77.9	66.5	53.9	43.6		
	5% DB	MDBR	11.5	15.2	18.8	20.1	21.2	21.6	22.6	23.9	22.2	19.1	12.3	11.0		
		MCDBR	16.2	23.0	28.5	28.7	26.6	25.9	27.0	27.9	28.3	27.5	20.1	16.8		
		MCWBR	11.3	15.9	15.6	14.9	11.0	9.0	8.4	8.9	10.2	13.0	12.0	12.8		
		MCWBR	13.8	22.1	26.7	26.3	24.6	24.3	23.8	24.6	23.7	24.5	18.3	17.0		
(40) Clear-Sky Solar Irradiance	taub		0.347	0.379	0.415	0.462	0.437	0.446	0.456	0.470	0.435	0.403	0.375	0.352		
		taud	2.275	2.204	2.155	2.043	2.192	2.204	2.181	2.137	2.201	2.271	2.303	2.292		
	Ebn at Noon		239	251	257	254	266	263	259	250	248	240	226	224		
		Edh at Noon	28	35	42	51	45	45	45	46	40	33	27	25		
(44) All-Sky Solar Radiation	RadAvg		459	718	1093	1504	1861	2035	2059	1835	1333	865	487	374		
	RadStd		69	100	110	153	147	140	135	110	103	102	66	55		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (55 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-230	N/A	N/A	

Nomenclature: See separate page